ON THE DESIGN OF BFFG

CAMILLA & FRANK (VU AMSTERDAM)

Contents

Abstract. Backward Filtering Forward Guiding is an algorithm for obtaining (weighted) samples of a partially observed stochastic process on a directed acyclic graph. For simplicity, we assume the state-space setting. The backward filtering step yields a sequence of *likelihood potentials* $\{g_i\}$ which are used in forward stochastic simulation of a *guided process* that is informed by these potentials. In this paper we investigate how properties of $\{g_i\}$ affect the efficiency of stochastic simulation of the guided process.

1. Introduction

We consider a latent discrete-time Markov process $(x_1, \dots, x_n)$. The edge from 0 to 1 fixes a prior on the initial state x_1. At times $i = 1, \dots, n$ we assume an observation $v_i \mid x_i$ is generated from a distribution with density $k(x_i, v_i)$. We aim to sample from $x_{1:n} \mid v_{1:n}$. Let $h_i(x_i)$ be the density of $v_{i:n} \mid x_i$. The Markov process that is conditioned on $v_{1:n}$ is a Markov process with transition densities

$$p^\star(x_i \mid x_{i-1}) = \frac{p(x_i \mid x_{i-1}) h_i(x_i)}{\int p(x_i \mid x_{i-1}) h_i(x_i) \, \mathrm{d}x_i}.$$

As the sequence of functions $\{h_i\}$ is often not available in closed form, we replace it with a sequence of *guiding* functions $\{g_i\}$. The resulting *guided* process has transition

Date: September 4, 2026.

densities

$$p^\circ(x_i \mid x_{i-1}) = \frac{p(x_i \mid x_{i-1}) g_i(x_i)}{\int p(x_i \mid x_{i-1}) g_i(x_i)\, \mathrm{d}x_i}$$

The smoothing density is given by

$$\begin{aligned} p^\star(x_{0:n}) &= \prod_{i=0}^n p^\star(x_i \mid x_{i-1}) = \prod_{i=0}^n \frac{p(x_i \mid x_{i-1}) h_i(x_i)}{\int p(x_i \mid x_{i-1}) h_i(x_i)\, \mathrm{d}x_i} \\ &= \frac{1}{h_0(x_0)} \left[\prod_{i=1}^n k(x_i, v_i)\right] \prod_{i=0}^n p(x_i \mid x_{i-1}), \end{aligned}$$

where $p(x_0 \mid x_{-1}) \equiv p(x_0)$. The final equality is shown in Theorem 6 in [vdMS25]. Under the guided process, we have

$$p^\circ(x_{0:n}) = \prod_{i=0}^n p^\circ(x_i \mid x_{i-1}) = \prod_{i=0}^n \frac{p(x_i \mid x_{i-1}) g_i(x_i)}{\int p(x_i \mid x_{i-1}) g_i(x_i)\, \mathrm{d}x_i},$$

Therefore

$$\frac{p^\star(x_{0:n})}{p^\circ(x_{0:n})} = \frac{1}{h_0(x_0)} \Psi(x_{0:n}), \tag{1}$$

where

$$\Psi(x_{0:n}) = \left[\prod_{i=1}^n k(x_i, v_i)\right] \prod_{i=0}^n \frac{\int p(x_i \mid x_{i-1}) g_i(x_i)\, \mathrm{d}x_i}{g_i(x_i)}.$$

In particular, the likelihood is given by

$$h_0(x_0) = \mathbb{E}^\circ \Psi(x_{0:n}). \tag{2}$$

1.1. **Research questions / goals.**

(1) Bound $\mathrm{KL}(\mathbb{P}^\star \| \mathbb{P}^\circ)$ by a discrepancy measure between $\{g_i\}$ and $\{h_i\}$. The behaviour of this KL-divergence characterises the quality of $\mathbb{P}^\circ$ as importance sampling distribution to sample from $\mathbb{P}^\star$ (Cf. [CD18]).

(2) The right-hand-side of (2) can be approximated by Monte-Carlo simulation from samples of the guided process. Under which conditions is this estimator of finite variance? If we have a candidate $\{g_i\}$ which is not guaranteed to have this property, can we fix this by transforming each g_i?

(3) As we can view the guided process as a proposal in a guided Feynman-Kac model, can we use general results from particle filtering to study the quality of the guided process?

2. Formulation as guided Feynman-Kac model

In the notation of Chapter 5 in [**?**], we have a guided Feynman-Kac model with

$$\begin{aligned} \mathbb{M}_0(\mathrm{d}x_0) &= p^\circ(x_0)\, \mathrm{d}x_0 \\ M_t(x_{t-1}, \mathrm{d}x_t) &= p^\circ(x_t \mid x_{t-1})\, \mathrm{d}x_t \\ f_t(v_t \mid x_t) &= k(x_t, v_t) \\ G_t(x_{t-1}, x_t) &= \frac{k(x_t, v_t) p(x_t \mid x_{t-1})}{p^\circ(x_t \mid x_{t-1})} = k(x_t, v_t) \frac{\int p(x_t \mid x_{t-1}) g_t(x_t)\, \mathrm{d}x_t}{g_t(x_t)}. \end{aligned}$$

In a generic particle filtering algorithm, Cf. Algorithm 10.1 in [**?**], the unnormalised weight is given by

$$w_t^n = G_t(X_{t-1}^{A_n}, X_t^n).$$

For particle filtering, by Proposition 11.3 in [**?**], if the potentials G_t are all upper bounded, then there exist constants c_t' such that

$$\mathbb{E}\left[\frac{1}{N}\sum_{n=1}^{N} W_t^n \varphi(X_t^n) - \mathbb{Q}_t(\varphi)\right)^2 \le c_t' \frac{\|\varphi\|_\infty^2}{N},$$

where $W_t^n = w_t^n / \sum_{m=1}^N w_t^m$.

The numerator of the expression for $G_t(x_{t-1}, x_t)$ is bounded provided k and g_t are, which is a reasonable assumption in applications. However, the denominator, $g_t(x_t)$, is typically not bounded away from zero. For that reason, we think of *transforming* the maps $\{g_t\}$ to ensure this. For example via

(A) mapping $g_t(x)$ to $g_t(x) + \epsilon_t$;
(B) mapping $g_t(x)$ to $\sqrt{g_t(x)^2 + \epsilon_t}$,

where $\{\varepsilon_t\}$ is a sequence of strictly positive numbers bounded away from zero. While such transformations on g_t are simple, for BFFG to be efficient, at the same time we need that

(1) sampling under p° is simple;
(2) evaluation of $\int p(x_t \mid x_{t-1}) g_t(x_t)\, \mathrm{d}x_t$ is simple. I think a positive unbiased estimator would suffice.

Option (A) ensures (1) as

$$\begin{aligned} p^\circ(x_t \mid x_{t-1}) &= \frac{p(x_t \mid x_{t-1})(g_t(x_t) + \epsilon_t)}{\int p(x_t \mid x_{t-1})(g_t(x_t) + \epsilon_t)\, \mathrm{d}x_t} \\ &= \lambda_t(x_{t-1}) \frac{p(x_t \mid x_{t-1}) g_t(x_t)}{\int p(x_t \mid x_{t-1}) g_t(x_t)\, \mathrm{d}x_t} + (1 - \lambda_t(x_{t-1})) p(x_t \mid x_{t-1}), \end{aligned}$$

where

$$\lambda_t(x_{t-1}) = \frac{\int p(x_t \mid x_{t-1}) g_t(x_t)\, \mathrm{d}x_t}{\int p(x_t \mid x_{t-1}) g_t(x_t)\, \mathrm{d}x_t + \epsilon_t}.$$

Hence, the guided process is a mixture distribution of the original guided process and the unconditional process. use rejection sampling for option (B)?

3. KL-bounds

Let $\mathbb{P}_T^*$ and $\mathbb{P}_T^\circ$ denote the laws of $X_{0:T}^*$ and $X_{0:T}^\circ$ respectively. According to results in [CD18], the quality of an importance sampling distribution, which is $\mathbb{P}_T^\circ$ here, is dictated by $\mathrm{KL}(\mathbb{P}_T^*, \mathbb{P}_T^\circ)$ given by

$$\mathrm{KL}\left(\mathbb{P}_T^* \| \mathbb{P}_T^\circ)\right) = \int p^*(x_{0:T}) \log\left(\frac{p^\star(x_{0:T})}{p^\circ(x_{0:T})}\right) dx_{0:T}\,.$$

We would like to bound this by a discrepancy measure between $\{h_i\}$ and $\{g_i\}$. [1] The following lemma is adapted from Proposition 3.1 in [JHJ21].

[1] We should keep in mind that the effect of h_i or g_i is invariant under multiplying with a positive constant.

Lemma 1. Define $\xi_t(x) = \log h_t(x) - \log g_t(x)$. Then

$$\log\left(\frac{p^\star(x_{0:T})}{p^\circ(x_{0:T})}\right) \leq \sum_{t=0}^{T} \xi_t(x_t) - \sum_{t=0}^{T} \int p^\circ(x_t \mid x_{t-1})\xi_t(x_t)dx_t$$

Proof. We have

$$\log\left(\frac{p^\star(x_{0:T})}{p^\circ(x_{0:T})}\right) = \log\left(\prod_{t=0}^{T} \frac{p^\star(x_t \mid x_{t-1})}{p^\circ(x_t \mid x_{t-1})}\right) = \log\left(\prod_{t=0}^{T} \frac{h_t(x_t)}{g_t(x_t)} \frac{\int p(x_t \mid x_{t-1})g_t(x_t)dx_t}{\int p(x_t \mid x_{t-1})h_t(x_t)dx_t}\right)$$
$$= \sum_{t=0}^{T} \log\left(\frac{h_t(x_t)}{g_t(x_t)}\right) + \sum_{t=0}^{T} \log\left(\frac{\int p(x_t \mid x_{t-1})g_t(x_t)dx_t}{\int p(x_t \mid x_{t-1})h_t(x_t)dx_t}\right)$$

using the convention $p^\star(x_0 \mid x_{-1}) = p(x_0)$ (and analogously for p°). Therefore, each summand in the second summation of satisfies:

$$\log\left(\frac{\int p(x_t \mid x_{t-1})g_t(x_t)dx_t}{\int p(x_t \mid x_{t-1})h_t(x_t)dx_t}\right) = \frac{\int p(x_t \mid x_{t-1})g_t(x_t)dx_t}{\int p(x_t \mid x_{t-1})g_t(x_t)dx_t} \cdot \log\left(\frac{\int p(x_t \mid x_{t-1})g_t(x_t)dx_t}{\int p(x_t \mid x_{t-1})h_t(x_t)dx_t}\right)$$
$$\leq \frac{1}{\int p(x_t \mid x_{t-1})g_t(x_t)dx_t} \int p(x_t \mid x_{t-1})g_t(x_t) \log\left(\frac{p(x_t \mid x_{t-1})g_t(x_t)}{p(x_t \mid x_{t-1})h_t(x_t)}\right) dx_t$$
$$= \int \frac{p(x_t \mid x_{t-1})g_t(x_t)}{\int p(x_t \mid x_{t-1})g_t(x_t)dx_t} \log\left(\frac{g_t(x_t)}{h_t(x_t)}\right) dx_t$$
$$= \int p^\circ(x_t \mid x_{t-1}) \log\left(\frac{g_t(x_t)}{h_t(x_t)}\right) dx_t\,,$$

where the inequality follows from the logsum-inequality given in Lemma 7. □

Corollary 2. If $c_t \leq \xi_t(x) \leq C_t$, then

$$\mathrm{KL}(\mathbb{P}_T^*, \mathbb{P}_T^\circ) \leq \sum_{t=0}^{T} (C_t - c_t).$$

The condition in this corollary is admittedly very strong.

Corollary 3. We have

$$\mathrm{KL}\left(\mathbb{P}_T^* \| \mathbb{P}_T^\circ)\right) \leq \sum_{t=0}^{T} \left(\mathbb{E}^\star \xi_t(X_t) - \sup_{x_{t-1}} \frac{p^\star(x_{t-1})}{p^\circ(x_{t-1})} \mathbb{E}^\circ\left[\min(\xi_t(X_t), 0)\right]\right), \tag{3}$$

where $p^\star(x_{t-1})$ and $p^\circ(x_{t-1})$ denote the marginal densities of x_{t-1} under $\mathbb{P}_T^\star$ and $\mathbb{P}_T^\circ$ respectively.

Proof. We have

$$\mathrm{KL}\left(\mathbb{P}_T^* \| \mathbb{P}_T^\circ)\right) \leq \sum_{t=0}^{T} \mathbb{E}^\star \xi_t(x_t) - \sum_{t=0}^{T} \int \int p^\circ(x_t \mid x_{t-1})\xi_t(x_t)\, \mathrm{d}x_t p^\star(x_{t-1}) dx_{t-1}.$$

The integral in the second term an be written as (divide and multiply by $p^\circ(x_{t-1})$ in the integral)

$$C := -\int \int \frac{p^\star(x_{t-1})}{p^\circ(x_{t-1})} p^\circ(x_{t-1}, x_t)\xi_t(x_t)\, \mathrm{d}x_t\, \mathrm{d}x_{t-1}.$$

Let

$$A = \{x_t \colon \xi_t(x_t) \geq 0\}.$$

We have

$$C \leq \int_{x_{t-1}} \int_{x_t \in A^c} p^\circ(x_{t-1}, x_t)(-\xi_t(x_t)) \, \mathrm{d}x_t \frac{p^\star(x_{t-1})}{p^\circ(x_{t-1})} \, \mathrm{d}x_{t-1}$$

as the integral will be negative when $x_t \in A$. Using Fubini, This leads to

$$\begin{aligned}
C &\leq \sup_{x_{t-1}} \frac{p^\star(x_{t-1})}{p^\circ(x_{t-1})} \int_{x_t \in A^c} p^\circ(x_t)(-\xi_t(x_t)) \, \mathrm{d}x_t \\
&= \sup_{x_{t-1}} \frac{p^\star(x_{t-1})}{p^\circ(x_{t-1})} \mathbb{E}^\circ \left[-\xi_t(X_t) \mathbf{1}_{\{-\xi_t(X_t) > 0\}} \right] \\
&= -\sup_{x_{t-1}} \frac{p^\star(x_{t-1})}{p^\circ(x_{t-1})} \mathbb{E}^\circ \left[\min(\xi_t(X_t), 0) \right].
\end{aligned}$$

In [JHJ21] it seems that $A = \emptyset$ is assumed. In that case, the term $\mathbb{E}^\circ \min(\xi_t(X_t), 0)$ can be replaced by $\mathbb{E}^\circ \xi_t(X_t)$.

□

In specific cases where $\mathbb{P}_T^\star$ is tractable, we may obtain sharper bounds.

Example 4. Gaussian case...

4. Argument by Ju/Heng

We start from Lemma 1.

By Fubini's theorem applied to the second term in Lemma 1, we get[2]

Corollary 5. If we define $p^{\star\circ}(x_0) = p^\circ(x_0)$ and for $t \geq 1$

$$p^{\star\circ}(x_t) = \int p^\star(x_{t-1}) p^\circ(x_t \mid x_{t-1}) \, \mathrm{d}x_{t-1},$$

then

$$\mathrm{KL}(\mathbb{P}_T^*, \mathbb{P}_T^\circ) \leq \sum_{t=0}^{T} \left(\mathbb{E}^\star \xi_t(X_t) - \mathbb{E}^{\star\circ} \xi_t(X_t) \right)$$

with $\mathbb{E}^{\star\circ}$ denoting expectation under $p^{\star\circ}$.

Then Heng defines $e_t = -\mathbb{E}^{\star\circ} \xi_t(X_t)$. In his particular application apparently $g_T = h_T$ which implies $e_T = 0$. We can write

$$e_t = \mathbb{E}^{\star\circ} \log \frac{g_t(X_t)}{h_t(X_t)} = \mathbb{E}^{\star\circ} \log \frac{g_t(X_t)}{\tilde{\psi}_t(X_t)} + \mathbb{E}^{\star\circ} \log \frac{\tilde{\psi}_t(X_t)}{h_t(X_t)},$$

where

$$\tilde{\psi}(x_t) = p(y_t \mid x_t) \int p(x_{t+1} \mid x_t) g_{t+1}(x_{t+1}) \, \mathrm{d}x_{t+1}.$$

[2]This is the first equation in the note by Jeremy Heng.

This is the BIF recursion, but with p rather than $\tilde{p}$ in the pullback. We have, using the logsum inequality

$$\begin{aligned}
\log \frac{\tilde{\psi}_t(x_t)}{h_t(x_t)} &= \log \frac{\int p(x_{t+1} \mid x_t) g_{t+1}(x_{t+1}) \,\mathrm{d}x_{t+1}}{\int p(x_{t+1} \mid x_t) h_{t+1}(x_{t+1}) \,\mathrm{d}x_{t+1}} \\
&\leq \frac{1}{\int p(x_{t+1} \mid x_t) g_{t+1}(x_{t+1}) \,\mathrm{d}x_{t+1}} \int p(x_{t+1} \mid x_t) g_{t+1}(x_{t+1}) \log \frac{g_{t+1}(x_{t+1})}{h_{t+1}(x_{t+1})} \,\mathrm{d}x_{t+1} \\
&= \int p^{\circ}(x_{t+1} \mid x_t) \log \frac{g_{t+1}(x_{t+1})}{h_{t+1}(x_{t+1})} \,\mathrm{d}x_{t+1}
\end{aligned}$$

This implies

$$\begin{aligned}
\mathbb{E}^{\star\circ} \log \frac{\tilde{\psi}(X_t)}{h_t(X_t)} &\leq \int \int p^{\circ}(x_{t+1} \mid x_t) \log \frac{g_{t+1}(x_{t+1})}{h_{t+1}(x_{t+1})} p^{\star\circ}(x_t) \,\mathrm{d}x_{t+1} \,\mathrm{d}x_t \\
&\leq M_t \int \int p^{\circ}(x_{t+1} \mid x_t) \log \frac{g_{t+1}(x_{t+1})}{h_{t+1}(x_{t+1})} p^{\star}(x_t) \,\mathrm{d}x_{t+1} \,\mathrm{d}x_t \\
&= M_t \int p^{\star\circ}(x_{t+1}) \log \frac{g_{t+1}(x_{t+1})}{h_{t+1}(x_{t+1})} \,\mathrm{d}x_{t+1} \\
&= M_t e_{t+1},
\end{aligned}$$

where

$$M_t = \sup_{x_t} \frac{p^{\star\circ}(x_t)}{p^{\star}(x_t)}.$$

Hence we obtain

$$e_t \leq \mathbb{E}^{\star\circ} \log \frac{g_t(X_t)}{h_t(X_t)} + M_t e_{t+1}, \tag{4}$$

to which we may apply Lemma 8. We conclude

Corollary 6.

$$\mathrm{KL}(\mathbb{P}_T^{*}, \mathbb{P}_T^{\circ}) \leq \sum_{t=0}^{T} \left(\mathbb{E}^{\star} \xi_t(X_t) + e_t\right)$$

where e_t satisfies the bound (4).

So what remains to be shown?

(1) Control of M_t (seems hard).

(2) Bounding $\mathbb{E}^{\star} \xi_t(X_t) = \mathbb{E}^{\star} \log \frac{h_t(X_t)}{g_t(X_t)}$. Cf. Equation (40) in Ju. I think here we have to repeat the whole argument with splitting up the term using $\tilde{\psi}$.

(3) Bounding $\mathbb{E}^{\star\circ} \log \frac{g_t(X_t)}{\tilde{\psi}_t(X_t)}$. Cf. arguments in Equations (86) and (87) in Ju (here, the logsum inequality is used).

Regarding (3), note that

$$\begin{aligned}
\log \frac{g_t(x_t)}{\tilde{\psi}_t(x_t)} &= \log \frac{\int \tilde{p}(x_{t+1} \mid x_t) g_{t+1}(x_{t+1}) \,\mathrm{d}x_{t+1}}{\int p(x_{t+1} \mid x_t) g_{t+1}(x_{t+1}) \,\mathrm{d}x_{t+1}} \\
&\leq \int \frac{\tilde{p}(x_{t+1} \mid x_t) g_{t+1}(x_{t+1})}{\tilde{p}(x_{t+1} \mid x_t) g_{t+1}(x_{t+1}) \,\mathrm{d}x_{t+1}} \log \frac{\tilde{p}(x_{t+1} \mid x_t)}{p(x_{t+1} \mid x_t)} \,\mathrm{d}x_{t+1}
\end{aligned}$$

by the logsum inequality. This term does not involve h and should therefore by tractable in specific cases/applications.

4.1. **On the proof in appendix B2 in [JHJ21].** This is about upper bounding $p^\star(x_t)/p^\circ(x_t)$. They argue that it suffices to bound $p^\star(x_{0:T})/p^\circ(x_{0:T})$. Why does this suffice? By (1) this boils down to bounding each of the terms in $\Psi(x_{0:n})$. Their argument, translated to our notation, is that h_t is bounded and then $\int p(x_t \mid x_{t-1} h_t(x_t \, \mathrm{d}x_t$ is bounded. Then, it suffices to lower bound $\int p(x_t \mid x_{t-1} g_t(x_t \, \mathrm{d}x_t$. Their proof for this is specific to their setting.

4.2. **On the proof in appendix B3 in [JHJ21].** This is about further bounding the bound in (3). In our notation, these bounds look like

$$\begin{aligned} \mathbb{E}^\star \log \xi_t(X_t) &\leq \sum_{k=t}^{T-1} c^\star_{k+1} \sqrt{\mathbb{E}^\star \varphi^\star(X_t)^2} \sqrt{\mathbb{E}^\star \Delta(X_t)} \\ -\mathbb{E}^\circ \log \xi_t(X_t) &\leq \sum_{k=t}^{T-1} c_{k+1} \sqrt{\mathbb{E}^\circ \varphi(X_t)^2} \sqrt{\mathbb{E}^\circ \Delta(X_t)}, \end{aligned}$$

where φ, $\varphi^\star$ and Δ are discrepancy measures of parameters in the true and approximate backward filters.

5. A toy example: Stochastic Volatility Model

Consider a state-space model where the observations are given by the random sequence

$$R_i = \exp\left(\frac{X_i}{2}\right) U_i, \quad 1 \leq i \leq n, \tag{5}$$

and the latent states by

$$X_i = \omega + \psi X_{i-1} + \eta W_i, \quad 1 \leq i \leq n, \tag{6}$$

where $(U_i)_{i=1}^n$, $(W_i)_{i=1}^n$ are independent random sequences of i.i.d. standard normal variates and X has initial distribution $X_0 \sim \mathcal{N}\left(\frac{\omega}{1-\psi}, \frac{\eta^2}{1-\psi^2}\right)$. We impose the parameter constraints $\omega \in \mathbb{R}$, $-1 < \psi < 1$ and $\eta > 0$.

We consider the following transformed version of the observation equation (5)

$$V_i \coloneqq \log\left(R_i^2\right) = X_i + U_i', \quad 1 \leq i \leq n, \tag{7}$$

where $U_i' \coloneqq \log\left(U_i^2\right)$, $i = 1, \ldots, n$.

5.1. **Backward Filtering.** The proposed model is simple in the sense that the latent process is linear Gaussian. If the distribution of U_i' would be Gaussian, the backward information filter provides a recursion that allows exact computation of $\{h_i\}$. This is unfortunately not the case, though can obtain guiding functions $\{g_i\}$ by backward filtering under the assumption

$$V_i \mid X_i \sim \mathcal{N}\left(\mu + X_i, \sigma^2\right) \tag{8}$$

with $\mu \coloneqq \mathbb{E}(U_i') \approx -1.27$ and $\sigma^2 \coloneqq \mathrm{Var}(U_i') = \pi^2/2$. In that case, the derivations are simple. It turns out to be useful to introduce the notation

$$U(c, F, H)(x) = \exp\left(c + Fx - \frac{1}{2}Hx^2\right). \tag{9}$$

It turns out that (8) implies any g_i can expressed in this form for a triplet (c_i, F_i, H_i).

- *Leaf Messages* From Theorem 6.1 (ii) in [vdMS25], if v_n is observed at a leaf, then $g_n(x) = U(c_n, F_n, H_n)(x)$ with $c_n = \log\left(\varphi(v_n; \mu, \sigma^2)\right)$, $F_n = (v_n - \mu)\sigma^{-2}$ and $H_n = \sigma^{-2}$.
- *Pullback Kernels* By statement (iii) of the same theorem, we have for the pullback $(\tilde{\kappa}_{n-1,n}, g_n)(x) = U\left(\bar{c}_n, \bar{F}_n, \bar{H}_n\right)(x)$ with

$$\begin{aligned}
\bar{H}_n &= \psi^2/C_n \\
\bar{F}_n &= \psi(F_n/H_n - \omega)C_n^{-1} \\
\bar{c}_n &= c_n - \log\left(\varphi^{\mathrm{can}}(0; F_n, H_n)\right) + \log\left(\varphi(\omega; F_n/H_n, C_n)\right),
\end{aligned}$$

  where $C_n = \eta^2 + H_n^{-1}$.

Here

$$k(x, v) = \exp(v - x)\, f_{\chi^2(1)}(\exp(v - x)).$$

This is bounded since for $u > 0$, the density of $u f_{\chi^2(1)}(u)$ behaves like $uu^{-1/2}e^{-u/2}$ which is bounded.

Appendix A. Some results

Lemma 7. Writing $a = \int a(x)dx$ and $b = \int b(x)dx$, the integral analogue of the log-sum inequality is

$$a \log\left(\frac{a}{b}\right) \le \int a(x) \log\left(\frac{a(x)}{b(x)}\right) dx,$$

Ssee e.g. its use in [CS04], Lemma 7.4.

Lemma 8. Let $(e_t)_{t=1}^T$, $(f_t)_{t=1}^T$, and $(M_t)_{t=1}^{T-1}$ be sequences of nonnegative real numbers such that

$$e_t \le f_t + M_t e_{t+1}, \qquad t = 1, \dots, T-1,$$

with terminal condition $e_T = 0$. Then

$$\sum_{t=1}^T e_t \le \sum_{k=1}^T \left(\sum_{t=1}^k \prod_{s=t}^{k-1} M_s\right) f_k,$$

where the empty product (when $t = k$) is taken to be 1.

Proof. We proceed by expanding the recursion. For $t = T-1$ we have

$$e_{T-1} \le f_{T-1} + M_{T-1}e_T = f_{T-1} + M_{T-1} \cdot 0 = f_{T-1}.$$

Inductively, suppose the claim holds for $e_{t+1}, \dots, e_T$. Then

$$e_t \le f_t + M_t e_{t+1}.$$

Expanding e_{t+1} gives

$$e_t \;\leq\; f_t + M_t f_{t+1} + M_t M_{t+1} f_{t+2} + \cdots + \Big(\prod_{s=t}^{T-1} M_s\Big) f_T.$$

Summing this inequality over $t = 1, \ldots, T$ yields

$$\sum_{t=1}^{T} e_t \;\leq\; \sum_{t=1}^{T} \sum_{k=t}^{T} \Big(\prod_{s=t}^{k-1} M_s\Big) f_k.$$

Reversing the order of summation shows that each f_k appears with coefficient

$$\sum_{t=1}^{k} \prod_{s=t}^{k-1} M_s,$$

giving the stated bound. □

Appendix B. Feynman-Kac models

B.1. Classical Feynman-Kac models (on chains). In the classical setting, a Feynman-Kac model describes a sequence of random variables $(X_0, X_1, \ldots, X_n)$ evolving according to:

- Markov transitions: $X_{k+1} \sim M_k(X_k, \cdot)$
- Potential functions: $G_k(x_{k-1}, x_k)$

The associated Feynman-Kac measure on paths is defined as:

$$\mathbb{Q}_n(dx_{0:n}) = \frac{1}{Z_n} \left[\prod_{k=0}^{n} G_k(x_{k-1}, x_k)\right] \mathbb{M}(dx_{0:n})$$

where $\mathbb{M}(dx_{0:n})$ is the law of the Markov chain, and Z_n is the normalizing constant.

Feynman-Kac models on directed trees. In a directed tree, the process evolves on a rooted tree T, where:

- Each node $t \in V(T)$ corresponds to a random variable X_t
- Each node t (except the root) has a parent $\mathrm{pa}(t)$, and a transition kernel $M_t(x_{\mathrm{pa}(t)}, \cdot)$
- Each node has a potential function $G_t(x_{\mathrm{pa}(t)}, x_t)$

The root r has an initial distribution: $X_r \sim \mu_r$

Definition. The Feynman-Kac measure $\mathbb{Q}_T$ on the state space $\{X_t : t \in T\}$ is given by:

$$\mathbb{Q}_T(\,\mathrm{d}\mathbf{x}) = \frac{1}{Z_T} \left[\prod_{t \in T} G_t(x_{\mathrm{pa}(t)}, x_t)\right] \mathbb{M}_T(\,\mathrm{d}\mathbf{x})$$

where $\mathbb{M}_T$ is the law of the Markov process over the tree:

$$\mathbb{M}_T(d\mathbf{x}) = \mu_r(\,\mathrm{d}x_r) \prod_{t \neq r} M_t(x_{\mathrm{pa}(t)},\, \mathrm{d}x_t)$$

and Z_T is the normalizing constant (partition function):

$$Z_T = \int \left[\prod_{t \in T} G_t(x_{\mathrm{pa}(t)}, x_t)\right] \mathbb{M}_T(\,\mathrm{d}\mathbf{x})$$

References

[CD18] Sourav Chatterjee and Persi Diaconis. The sample size required in importance sampling. *Annals of Applied Probability*, 28(2):1099–1135, 2018.

[CS04] Imre Csiszár and Paul C Shields. *Information theory and statistics: A tutorial*. Now Publishers, Inc., 2004.

[JHJ21] Nianqiao Ju, Jeremy Heng, and Pierre E. Jacob. Sequential monte carlo algorithms for agent-based models of disease transmission. *arXiv preprint arXiv:2101.12156*, 2021.

[vdMS25] Frank H. van der Meulen and S. Sommer. Backward filtering forward guiding. *Journal of Machine Learning Research*, 26(281):1–51, 2025.